

HAZMAT TRANSPORTATION REQUIREMENTS

Training, Packaging, and Shipping Paper Guide for Transporting Hazmat

U.S. Department of Transportation

Pipeline and Hazardous Materials Safety Administration

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HAZMAT TRAINING REQUIREMENTS

The Federal hazardous materials transportation law (49 U.S.C. 5101 et seq.) is the statute that pertains to the transportation of hazardous materials (hazmat) in the United States, and that requires the training of ALL hazmat employees. The purpose of this training is to increase a hazmat employee's safety awareness and is an essential element in reducing hazmat incidents. The Hazardous Materials Regulations (HMR) includes training requirements in several sections of Title 49 Code of Federal Regulations (CFR) as follows:

- GENERAL §173.1
- SPECIFIC §172.704
- MODAL
  - ◊◊ Air §175.20
  - ◊◊ Vessel §176.13
  - ◊◊ Highway §§177.800, 177.816

Receiving the required training enhances employee safety and security and increases employee productivity and skills. Effective training also reduces incidents and accidents thereby reducing operating costs and losses from property damage, thus increasing profits.

HMR TRAINING REQUIREMENTS

Each hazmat employer must train and test their hazmat employees, certify their training, and develop and retain records of current training.

Hazmat training must include:

- general awareness/familiarization;
- function-specific;
- safety;
- security awareness;
- in-depth security training, if a security plan is required; and
- driver training (for each hazmat employee who will operate a motor vehicle).

FREQUENCY OF TRAINING

Initial training of new hazmat employees, or an employee who changes job functions, must be completed within 90 days of employment or change in job function. A new employee may perform hazmat job functions before completing training provided the employee does so under the direct supervision of a properly trained and knowledgeable hazmat employee.

Recurrent training is required at least once every three years. The three-year period begins on the actual date of training. Relevant training received from a previous employer or source may be used to satisfy the requirements provided a current record of training is obtained from the previous employer or other sources.

Training conducted by OSHA, EPA, and other Federal or international agencies may be used to satisfy the training requirements in §172.704(a) to the extent that such training addresses the components specified (general awareness/familiarization; function-specific; safety; security awareness; in-depth security training, if a security plan is required; and driver training for each hazmat employee who will operate a motor vehicle).

TRAINING RECORDS

Training records must be kept by the hazmat employer for each hazmat employee, and must include the following:

- the hazmat employee's name;
- the completion date of the most recent training;
- training materials used (copy, description, or location);
- the name and address of the hazmat trainer; and
- certification that the hazmat employee has been trained and tested.

Training records must be retained for each hazmat employee for three years from the date of the last training, and for 90 days after the employee leaves the company or their position.

DEFINITIONS

Training - a systematic program (consistent approach, testing, and documentation) that ensures a hazmat employee has knowledge of hazmat and the HMR, and can perform assigned hazmat functions properly. See §172.700 through §172.704.

Hazmat employer - a person who employs one or more employees and who:

- transports hazmat in commerce;
- causes hazmat to be transported or shipped in commerce; or
- designs, manufactures, fabricates, inspects, represents, marks, certifies, sells, offers, reconditions, tests, repairs, or modifies packagings as qualified for use in the transportation of hazmat.

The term "hazmat employer" also includes any department, agency, or instrumentality of the United States, a State, a political subdivision of a State, or Native American Indian tribe engaged in offering or transporting hazmat in commerce. This term includes a person who is self-employed, including an owner-operator of a motor vehicle that transports hazmat in commerce.

Hazmat employee - a person employed by a hazmat employer, or person who is self-employed, and who directly affects hazmat transportation safety including:

- an owner-operator of a motor vehicle that transports hazmat;
- a person who:
  - ◊ Loads, unloads, or handles hazmat;
  - ◊ designs, manufactures, fabricates, inspects, tests, reconditions, repairs, modifies, marks, or otherwise represents packagings as qualified for use in the transportation of hazmat;
  - ◊ prepares hazmat for transportation;
  - ◊ is responsible for safety of transporting hazmat; or
  - ◊ operates a vehicle used to transport hazmat.

Compliance Check: Refer to §171.8 for definitions.

PREPARING SHIPPING PAPERS

§§172.200 - 205, 172.602, 172.604

The proper shipping description of a hazardous material consists of:

- a Basic Description,
- additional information (depending on materials and mode of transport),
- quantity of hazardous material, and
- Number and type of packaging used.

The Basic Decription of a hazardous material includes the Identification number, the Proper Shipping Name, Hazard Class, and Packing Group (when applicable). This information must be placed on the shipping paper in the specific order required in Part 172, Subpart C of the HMR. The HMR do not require a shipper to use a special form or format—it requires the proper information be placed on the shipping papers in the proper sequence.

SHIPPING DESCRIPTION

A correct shipping description includes many components, including the Basic Description, which must be placed on a shipping paper in the sequence required in §172.202(b) of the HMR. An easy way to remember this sequence is to refer to the acronym "ISHP": **I**dentification Number, **S**hipping Name, **H**azard Class or Division, and **P**acking Group. If a technical name is required, it must be placed in parentheses and listed after the Proper Shipping Name or Basic Description. If applicable, a subsidiary hazard class(es) must be placed in parentheses immediately following the primary hazard class.

The total amount of hazardous materials covered by each description must be indicated by mass or volume with the applicable unit of measure. For example: "200 kgs" or "50 L." The number and type of packages must be indicated and may include the packaging specification, for example: "12 drums," "12 1H1 drums," or "12 drums (UN 1A1)." The total quantity and types of packagings must be entered before, after, or both before and after the Basic Description.

ADDITIONAL DESCRIPTION

When additional information is required or provided it must be listed after the Basic Description, unless the HMR states otherwise. Looking at the "Check List" in this guide, you will see many of these identified under Additional Descriptions. Always check §172.203 of the HMR to ensure you have identified all the requirements that pertain to your shipment.

Two exceptions to the regulation for placing additional information after the Basic Description pertain to the "Technical name" and the letters "RQ." The "Technical Name" may be placed in parentheses after the Proper Shipping Name or after the Basic Description. "RQ" may be entered either before or after the Basic Description. On a shipping paper with columns specifically identifying hazardous materials, "RQ" may replace the "X" normally placed in that column. Some hazardous materials, such as radioactive materials, require much more specific information than is covered here. Always use Part 172, Subpart C, of the HMR for specific details and other information pertaining to your shipment.

CHECKLIST

Check the HMR, Part 172, Subpart C, for specific Shipping paper requirements that may pertain to the materials you are shipping. Use the following as a guide.

The acronym "ISHP" spells out the proper sequence of the Basic Description, as shown: Basic Description (proper sequence §172.202(b)). Use the Hazardous Materials Table (HMT) (§172.101).

Identification Number (HMT Column 4); Proper Shipping Name (HMT Column 2); Hazard Class (HMT Column 3); Packing Group (HMT Column 5);

Other Information (§172.202)

① Technical name ("G" in HMT Column 1). See HMR §172.203(k) for specific requirements.
② Subsidiary hazard(s) (HMT Column 6)
③ Number and type of packages (i.e., "12 drums"or "12 1A1 drums")
④ Total quantity (by mass or volume, i.e., "200 kgs"or "50 L")
⑤ "Net Explosive Mass" for Class 1. See HMR §172.202(a)(5)(i) for specific requirements.
⑥ "EX-xxxx" for Class 1 materials when required. See HMR §172.320(d) for specific requirements.

Additional Descriptions (§172.203)

"DOT-SPxxxxx" Special Permit Number
"Ltd Qty" or "Limited Quantity"
⑦ "RQ" for Reportable Quantity. See Appendix A to §172.101 for specific requirements.
"RESIDUE: LAST CONTAINED***" (Required for tank car with residue)
⑧ "Marine Pollutant" (for non-bulk by vessel and bulk in all modes). See Appendix B of the HMT for specific requirements.
⑨ "Poison-Inhalation Hazard" or "Toxic-Inhalation Hazard" and applicable Hazard Zone, e.g., "Zone A," "Zone B," etc., (HMT Column 7 and §172.102)
"HOT" for liquid elevated temperature materials, when molten or elevated temperature is not part of proper shipping name. See HMR §172.203(n) for specific requirements.
"Organic Peroxides" See HMR §172.203(o) for specific requirements.
"Waste" See HMR §172.102 for specific requirements.
"Radioactive Material" See HMR §172.203(d) for specific requirements.
⑩ Mode Requirements - Additional information may be required depending on specific mode(s) of transport. See HMR §172.203 for specific requirements.

Emergency Response Telephone Number (§172.201(d), §172.604)

⑪ "EMERGENCY CONTACT: xxx-xxx-xxxx" (include international access code, if applicable). Offeror's name must appear directly before, after, above, or below the ER number unless clearly visible elsewhere.
⑫ If the ER number has been contracted with a capable agency, you must include the offeror's name or the contract number.

Shipper's Certification (§172.204)

⑬ Signed statement: "This is to certify that the above mentioned materials..." See HMR §172.204 for specific certification statements.

USING THE HAZARDOUS MATERIALS TABLE (HMT) §172.101

Symbols (1)	Hazardous Materials Descriptions and Proper Shipping Names (2)	Hazard Class or Division (3)	Identification Numbers (4)	PG (5)	Label Codes (6)	Special Provisions (§172.102) (7)	(8) Packaging (§173.***) Exceptions (8A)	Non-bulk (8B)	Bulk (8C)	(9) Quantity Limitations Passenger aircraft/rail (9A)	Cargo aircraft only (9B)
	Acrolein, stabilized	6.1	UN1092	I	6.1, 3	1, B9, B14, B30, B42, B77, T22, TP2, TP7, TP13, TP38, TP44	None	226	244	Forbidden	Forbidden
	Copper cyanide	6.1	UN1587	II	6.1	IB8, IP2, IP4 T3, TP33	153	204	242	25kg	100kg
G	Flammable liquids, n.o.s	3	UN1993	I II III	3 3 3	T11, TP1, TP27 IB2, T7, TP1,TP8, TP28 B1, B52, IB3, T4, TP1, TP29	150 150 150	201 202 203	243 242 242	1L 5L 60L	30L 60L 220L
	Phosphoric acid solution	8	UN1805	III	8	A7, IB3, N34, T4, TP1	154	203	241	5L	60L

The HMT (§172.101) provides a list of hazardous materials. The first step in filling out a shipping paper correctly is to refer to the HMT and find the entry that most appropriately describes the hazardous material(s) you are shipping. This information, known as the Basic Description, will include the Identification Number in Column 4, Proper Shipping Name in Column 2, Hazard Class or Division in Column 3, and Packing Group in Column 5. If a material has one or more subsidiary hazards, they are identified in Column 6. Subsidiary hazards must be listed with the basic description.

Codes listed in Columns 1 and 7 indicate there is additional information regarding the hazardous material being shipped—some of which may be required to be entered with the basic description. For example, a "G" in Column 1 indicates that the proper shipping name listed must be further identified by the addition of a "technical name" placed in parentheses. The chemical manufacturer or the safety data sheet should provide this information. Special provision codes listed in Column 7 are defined in §172.102. These codes can indicate many things, including whether a material is toxic by inhalation.

After the material is identified in the HMT, Appendices A and B must be checked. Appendix A is the List of Hazardous Substances and Reportable Quantities (RQ). If the material is listed in Appendix A, you must determine if it meets the definition of a hazardous substance(s) prescribed in §171.8. If the material is listed, and the quantity per package meets or exceeds the RQ amount listed, identify it as a reportable quantity on the shipping paper. (See HMR §172.203(c) for specific requirements.

Appendix B lists marine pollutants. Most packages of hazardous materials that meet the definition of a marine pollutant must be identified as such when transported by vessel. When transported by modes other than vessel, only marine pollutant bulk packages must be identified as such. (See HMR §172.203(l) for specific requirements and §171.4 for exceptions).

DESCRIPCIONES DE ENVIO

Hazardous and Non-hazardous Materials*

No. of Units & Container Type ③	HM	BASIC DESCRIPTION Identification Number (UN or NA), Proper Shipping Name, Hazard Class, Packing Group, per 172.101, 172.202, 172.203	TOTAL QUANTITY (Weight, Volume, Gallons, etc.) ④
1 Box		Carriage Bolts	1000 lbs
4 Drums	X	UN1805, Phosphoric acid solution, 8, PGIII	4 gal
1 Drum	X	UN1993, Flammable liquids, n.o.s.	18 gal
		(contains methanol), 3, PGIII ①	
		⑩ This shipment is within limitations for cargo aircraft only.	

*When not listed first, use a contrasting color or highlight the **Basic Description**, and/or place an "X" in the "HM" column when provided.

Toxic Inhalation and Subsidiary Risk

No. of Units & Container Type	BASIC DESCRIPTION Identification Number (UN or NA), Proper Shipping Name, Hazard Class, Packing Group, per 172.101, 172.202, 172.203	TOTAL QUANTITY (Weight, Volume, Gallons, etc.)
③ 10 drums	② UN1092, Acrolein, stabilized, 6.1(3), PGI, Toxic-Inhalation Hazard, Zone A ⑨	④ 1 gal

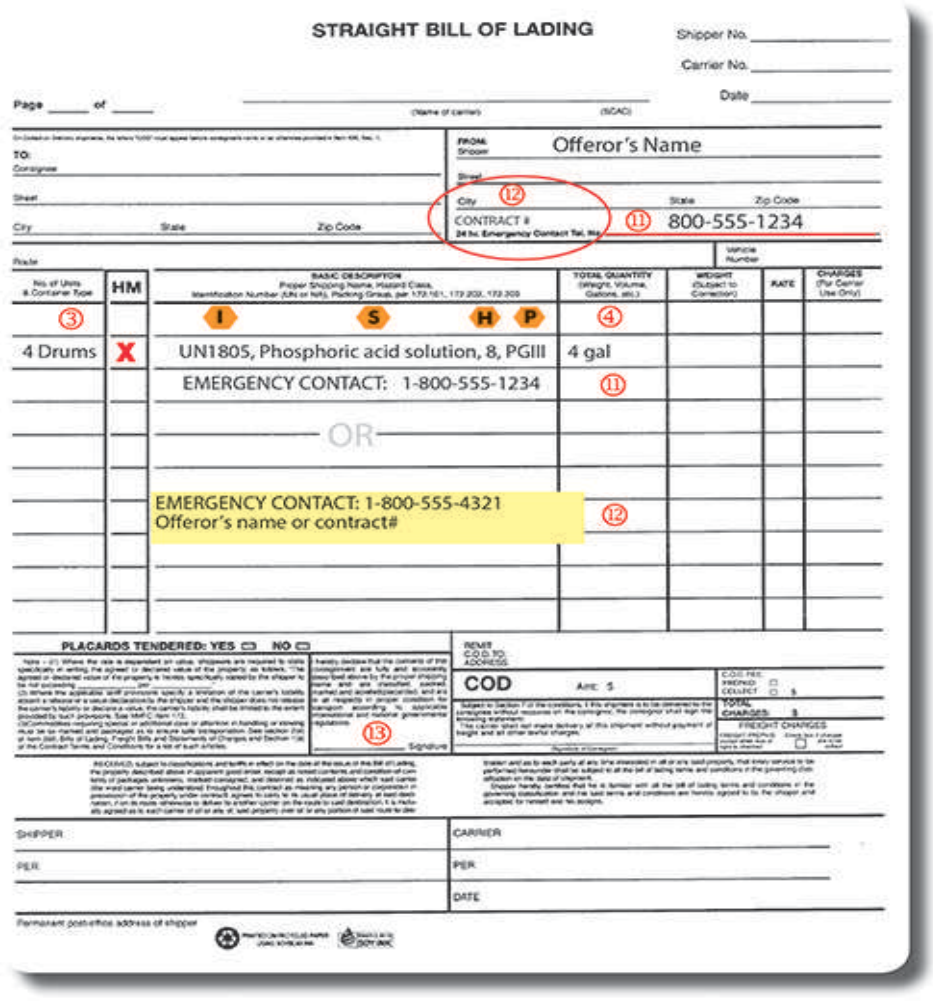
Reportable Quantity and Marine Pollutant

No. of Units & Container Type	HM	BASIC DESCRIPTION Identification Number (UN or NA), Proper Shipping Name, Hazard Class, Packing Group, per 172.101, 172.202, 172.203	TOTAL QUANTITY (Weight, Volume, Gallons, etc.)
③		⑦	④
1 Box	X	RQ, UN1587, Copper Cyanide, 6.1, PGII, Marine Pollutant	10 lbs
	⑦	⑧ OR	
1 Box	RQ	UN1587, Copper Cyanide, 6.1, PGII, Marine Pollutant	10 lbs
		⑧	

Explosives

No. of Units & Container Type	BASIC DESCRIPTION Identification Number (UN or NA), Proper Shipping Name, Hazard Class, Packing Group, per 172.101, 172.202, 172.203	TOTAL QUANTITY (Weight, Volume, Gallons, etc.)
③ 1 Box	UN0030, Detonators, electric, 1.1B EX-number (as applicable) ⑥	⑤ .008 Net Explosive Mass

SAMPLE SHIPPING PAPER



STRAIGHT BILL OF LADING

Shipper No. _____

Carrier No. _____

Date _____

Page ____ of ____

FROM Shipper: Offeror's Name

Street

City ⑫ State Zip Code

CONTRACT # ⑪ 800-555-1234

24 hr. Emergency Contact Tel. No.

TO: Consignee

Street

City State Zip Code

Route

No. of Units & Container Type	HM	BASIC DESCRIPTION Proper Shipping Name, Hazard Class, Identification Number (UN or NA), Packing Group, per 172.101, 172.202, 172.203	TOTAL QUANTITY (Weight, Volume, Gallons, etc.)	WEIGHT (Subject to Correction)	RATE	CHARGES (For Carrier Use Only)
③		I S H P	④			
4 Drums	X	UN1805, Phosphoric acid solution, 8, PGIII	4 gal			
		EMERGENCY CONTACT: 1-800-555-1234	⑪			
		OR				
		EMERGENCY CONTACT: 1-800-555-4321 Offeror's name or contract#	⑫			

PLACARDS TENDERED: YES ☐ NO ☐

REMIT C.O.D. TO: ADDRESS

COD Amt: $

C.O.D. FEE PREPAID ☐ COLLECT ☐ $

TOTAL CHARGES: $

FREIGHT CHARGES

⑬ Signature

SHIPPER

PER

DATE

CARRIER

PER

DATE

Permanent post-office address of shipper

SHIPPER'S CERTIFICATION

Persons offering hazardous materials for transportation shall certify that their shipment is offered in accordance with the HMR. The required certification statements are listed in § 172.204. The certification must be signed by a principal, officer, partner, or employee of the shipper or his agent. It may be signed manually, by typewriter, or by other mechanical means.

RETENTION

The person providing the shipping paper shall record the date the hazardous material is accepted by the carrier (this may vary for rail, vessel, or air) and maintain a copy or electronic image of the shipping paper for two years (three years for a hazardous waste). The copies must be accessible at or through the principal place of business and must be made available if requested by an authorized official of a Federal, State, or local government agency at reasonable times and locations. See §172.201 for specific requirements.

EMERGENCY RESPONSE INFORMATION

Shipments of hazardous materials must include emergency response information on the shipping paper itself or a separate document. This emergency response information is used in the event of an incident involving hazardous materials to assist emergency responders at the scene. At a minimum, this information must include: (1) the Basic Description (including technical name, if applicable), (2) immediate hazards to health, (3) risks of fire or explosion, (4) immediate precautions to be taken in the event of an accident or incident, (5) immediate methods for handling fires, (6) initial methods for handling spills or leaks in the absence of fire, and (7) preliminary first aid measures. For additional and specific requirements, check Part 172, Subpart G of the HMR.

EMERGENECY RESPONSE TELEPHONE NUMBER

Shipping papers must contain an emergency response telephone number unless specifically excepted in §172.604. This number must include the area code or international access code, as applicable, and be monitored at all times while the material is in transit or in storage incidental to transportation. The person answering must be knowledgeable of the hazardous materials being shipped, or have immediate access to a person with such knowledge. The person must be able to assist first responders at the scene of an incident involving the hazardous materials (e.g., fire or explosion hazards, protective clothing required, and evacuation distances). Answering services, answering machines, beepers, or telephone numbers that require a call back are not permitted.

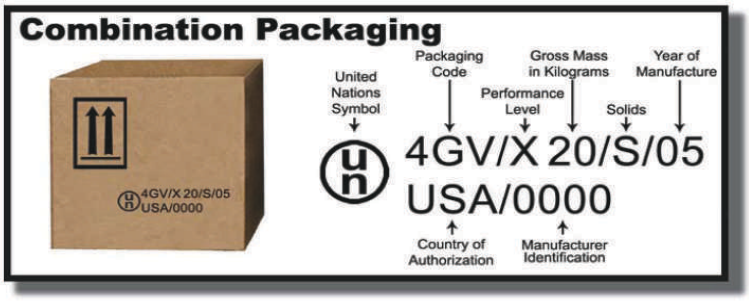
The emergency response telephone number must be entered immediately following the shipping description for the material or entered once in a prominent area on the shipping paper that is clearly visible and easily identified, for example: "EMERGENCY CONTACT: XXX-XXX-XXXX." If the number is placed in one location, it must apply to all the hazardous materials described on the shipping paper. If separate telephone numbers are needed for different materials, the corresponding emergency response number must be entered immediately after the shipping description to which it applies.

The telephone number may be the number of the person offering the shipment or the number of a hired third-party provider as long as the person monitoring it has the most current information from the offeror on the material and accepts responsibility for providing this information in an emergency. Note that many third-party providers require a contract/fee to provide this service. If a third-party emergency response provider is used, the person who is registered with the provider must be identified by name, contract number, or other unique identifier on the shipping paper directly before, after, above, or below the emergency response number. This information can be entered elsewhere on the shipping paper provided it is prominent, clearly visible, and allows the information to be easily and quickly found.

PERFORMANCE PACKAGING CODES

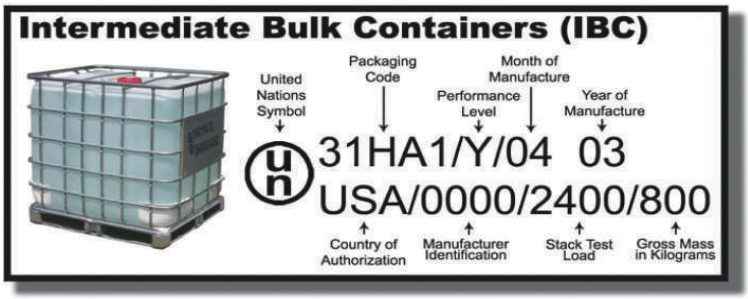
PACKING GROUP (PG)

Packing group means a grouping according to the degree of danger presented by hazardous materials. Packing Group I indicates great danger; Packing Group II, medium danger; Packing Group III, minor danger. Class 1, Class 2, Class 7, Division 6.2 (other than regulated medical wastes), articles (e.g., lithium batteries), and ORM-D materials do not have packing groups.









PACKAGING CODES

§§178.504 - 178.521	
1A1	Steel drum, non-removable head
1A2	Steel drum, removable head
1B1	Aluminum drum, non-removable head
1B2	Aluminum drum, removable head
1D	Plywood drum
1G	Fiber drum
1H1	Plastic drum, non-removable head
1H2	Plastic drum, removable head
1N1	Metal drum, non-removable head
1N2	Metal drum, removable head
2C1	Wooden barrel, bung type
2C2	Wooden barrel, slack type, removable head
3A1	Steel jerrican, non-removable head
3A2	Steel jerrican, removable head
3B1	Aluminum jerrican, non-removable head
3B2	Aluminum jerrican, removable head
3H1	Plastic jerrican, non-removable head
3H2	Plastic jerrican, removable head
4A	Steel box
4B	Aluminum box
4N	Metal box, other
4C1	Wood box, ordinary
4C2	Wood box, sift-proof walls
4D	Plywood box
4F	Reconstituted wood box
4G	Fiberboard box
4H1	Plastic box, expanded
4H2	Plastic box, solid
5H1	Woven plastic bag, unlined or non-coated
5H2	Woven plastic bag, sift-proof
5H3	Woven plastic bag, water-resistant
5H4	Plastic film bag
5L1	Textile bag, unlined or non-coated
5L2	Textile bag, sift-proof
5L3	Textile bag, water-resistant
5M1	Paper bag, multi-wall
5M2	Paper bag, multi-wall water-resistant
§§178.522 and 178.523	
6HA1	Plastic receptacle within a protective steel drum
6HA2	Plastic receptacle within a protective steel crate or box
6HB1	Plastic receptacle within a protective aluminum drum
6HB2	Plastic receptacle within a protective aluminum crate or box
6HC	Plastic receptacle within a protective wooden box
6HD1	Plastic receptacle within a protective plywood drum
6HD2	Plastic receptacle within a protective plywood box
6HG1	Plastic receptacle within a protective fiber drum
6HG2	Plastic receptacle within a protective fiberboard box
6HH1	Plastic receptacle within a protective plastic drum
6HH2	Plastic receptacle within a protective plastic box
6PA1	Glass, porcelain, or stoneware receptacles within a protective steel drum
6PA2	Glass, porcelain, or stoneware receptacles within a protective steel crate or box
6PB1	Glass, porcelain, or stoneware receptacles within a protective aluminum drum
6PB2	Glass, porcelain, or stoneware receptacles within a protective aluminum crate or box
6PC	Glass, porcelain, or stoneware receptacles within a protective wooden box
6PD1	Glass, porcelain, or stoneware receptacles within a protective plywood drum
6PD2	Glass, porcelain, or stoneware receptacles within a protective wickerwork hamper
6PG1	Glass, porcelain, or stoneware receptacles within a protective fiber drum
6PG2	Glass, porcelain, or stoneware receptacles within a protective fiberboard box
6PH1	Glass, porcelain, or stoneware receptacles within a protective expanded plastic packaging
6PH2	Glass, porcelain, or stoneware receptacles within a protective solid plastic packaging

INTERPRETING MARKINGS §§178.502 and 178.503

United Nations Symbol: For embossing metal receptacles, the letters UN may be applied in place of the symbol.

Packaging Code: Designates the type of packaging and material of construction.

A letter "W" designates associate administrator approval. A letter "V" designates special "variation" packaging.

Performance Level: Identifies the performance standard for successful testing of the packaging.

- **X -** For packagings meeting Packing Group I, II, and III tests.
- **Y -** For packagings meeting Packing Group II and III tests.
- **Z -** For packagings meeting Packing Group III tests.

Specific Gravity: Specific gravity for which the packaging design type has been tested. If the specific gravity does not exceed 1.2, the designation may be omitted.

Gross Mass: Maximum permitted gross mass in kilograms for the packaging design type.

S: Designates that the packaging is intended to contain solids or inner packagings.

Hydrostatic Test Pressure: Internal Hydrostatic Test Pressure in Kilopascals. This test is not required for inner packaging of combination packaging.

Year of Manufacture: Last two digits of year. of manufacture. Plastic drums and jerricans (1H and 3H) must be marked with the month of manufacture. The month marking may be located elsewhere on the package.

Country of Authorization: Country designation code indicates where the package is manufactured and marked.

Manufacturer Identification: Name and address or authorized symbol of packaging manufacturer or certifying agency.

Minimum Thickness: For metal or plastic drums, jerricans or the outer packaging of composite packaging intended for reuse or reconditioning.

R: Reconditioned packaging.

L: Reconditioned packaging having successfully passed a leakproofness test. This test is not required for inner packaging of combination packaging.

Note: Additional marking requirements for Reconditioned packaging are found in §178.503.

Additional Requirements for Intermediate Bulk Containers (IBC).

Month and Year: For IBCs, the month and year of manufacture.

Stacking Test Load: For IBCs, the stacked test load in kilograms. A number "0" shows that the IBC is not designed for stacking.

Note: Standards and codes for IBCs are found in §178.702 - 178.710.

REQUIRED TESTS FOR NON-BULK PACKAGING §§178.600-178.609

Drop Test (§178.603) All packaging design types.

Leakproofness Test (§178.604) All packaging design types intended for liquids.

Hydrostatic Test (§178.605) All metal, plastic, and composite design types intended to contain liquids.

Stacking Test (§178.606) All packaging design types other than bags.

Cooperage Test (§178.607) All bung-type wooden barrels.

Vibration Test (§178.608) All packaging design types.

Infectious Substances See §178.609 for test requirements for packagings for infectious substances.

Pressure Differential (§173.27) Packagings intended for air transport.

CONVERSION FACTORS §171.10

For compatibility with international transportation standards, most units of measure used in Subchapter C of the 49 CFR are expressed using International System of Units (SI). Where SI units appear (L, kg, kPa) they are the regulatory standard.

U.S. standard or customary units, (gal., lbs., psi) appearing in parentheses following SI units are for information only and not intended to be the regulatory standard.

10 kPa = 1.45 psi

75 kPa = 10.9 psi

95 kPa = 13.8 psi

100 kPa = 14.5 psi

150 kPa = 21.8 psi

250 kPa = 36.3 psi

1 kg = 2.2 lb

25 kg = 55 lb

50 kg = 110 lb

100 kg = 220 lb

150 kg = 330 lb

200 kg = 440 lb

250 kg = 551 lb

0.47 L = 1 pint (US)

0.95 L = 1 quart (US)

1 L = .264 gal (US)

3.8 L = 1 gal (US)

4 L = 1.06 gal (US)

5 L = 1.32 gal (US)

10 L = 2.6 gal (US)

CONVERSION TABLE
1 lb. = .454 kg; 1 gal. = 3.785 L

Compliance Check. Use the 49 CFR, Parts 100-185.

FEATURED RESOURCES

1. **Hazmat Transportation Training Modules** Online
2. **Check the Box Undeclared Hazmat Shipments Awareness Campaign Materials**
3. **DOT CHART 16** Publication
4. **DOT CHART 16** Mobile App
5. **Hazmat Registration Search** Mobile App
6. **Hazmat Transportation Requirements** Publication
7. **How to Use the HMR** Publication



All Mobile Apps available at

Download on the App Store | GET IT ON Google play

For additional information contact:

The Hazardous Materials Info Center

1-800-HMR-4922

(1-800-467-4922)

Email: infocntr@dot.gov

http://hazmat.dot.gov

Pipeline and Hazardous Materials Safety Administration

Outreach, Training, and Grants Division

East Building, 2nd Floor

1200 New Jersey Ave., SE

Washington, DC 20590

Email: training@dot.gov

202-366-4900

202-366-7342 (Fax)

U.S. Department of Transportation

Pipeline and Hazardous Materials Safety Administration

PHH50-0184-0220

CHECK THE BOX

checkthebox.dot.gov

U.S. Department of Transportation

Pipeline and Hazardous Materials Safety Administration

DOT CHART 17

Hazardous Materials Markings, Labeling and Placarding Guide

Refer to 49 CFR, Part 172:

Marking - Subpart D

Labeling - Subpart E

Placarding - Subpart F



NOTE: This document is for general guidance only and should not be used to determine compliance with 49 CFR, Parts 100-185.

HAZARDOUS MATERIALS MARKINGS

Package Orientation (Red or Black)



§172.312(a)

Keep Away from Heat



§172.317

Overpack Mark



§173.25(a)(4)

Fumigant Marking



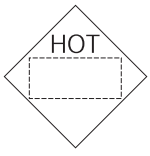
§172.302(g) and §173.9

Material Poisonous by Inhalation



§172.313(a)

Elevated Temperature Material



§172.325

UN ID Number Mark



§172.332(a)

Biological Substances, Category B



§173.199 (a)(5)

Lithium Battery Handling Mark



§173.185

Marine Pollutant



§172.322

Limited Quantity



§172.315

Biohazard Mark



§172.323

Petroleum Sour Crude Oil



§172.327

Excepted Quantity



§173.4a(g)

Hazardous Materials Warning Labels

Actual label size: at least 100 mm (3.9 inches) on all sides

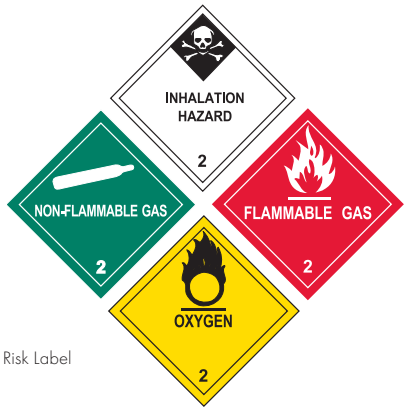
CLASS 1 Explosives: Divisions 1.1, 1.2, 1.3, 1.4, 1.5, 1.6



Subsidiary Risk Label

§172.411

CLASS 2 Gases: Divisions 2.1, 2.2, 2.3



§172.405(b), §172.415, §172.416, §172.417

CLASS 3 Flammable Liquid



§172.419

CLASS 4 Flammable Solid, Spontaneously Combustible, and Dangerous When Wet: Divisions 4.1, 4.2, 4.3



§172.420, §172.422, §172.423

CLASS 5 Oxidizer, Organic Peroxide: Divisions 5.1 and 5.2



§172.426, §172.427

* Include compatibility group letter.

** Include division number and compatibility group letter.

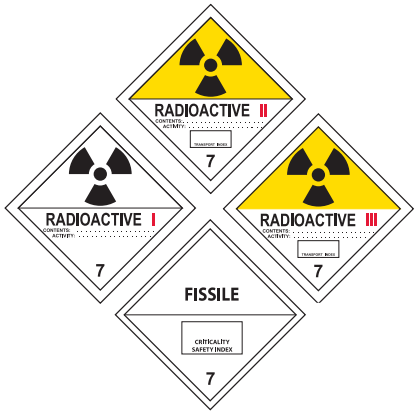
CLASS 6 Poison (Toxic), Poison Inhalation Hazard, Infectious Substance: Divisions 6.1 and 6.2



§172.323, §172.405(c), §172.429, §172.430, §172.432

For Regulated Medical Waste (RMW), an Infectious Substance label is not required on an outer packaging if the OSHA Biohazard marking is used as prescribed in 29 CFR 1910.1030(g). A bulk package of RMW must display a BIOHAZARD marking.

CLASS 7 Radioactive



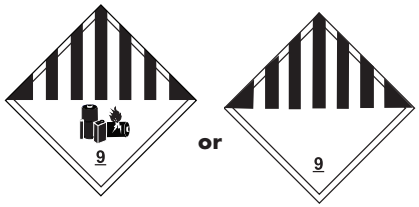
§172.436, §172.438, §172.440, §172.441

CLASS 8 Corrosive



§172.442

CLASS 9 Miscellaneous Hazardous Material



§§172.446, §172.447

Cargo Aircraft Only



§172.448

Empty Label



§172.450

Hazardous Materials Warning Placards

Actual placard size: at least 250 mm (9.84 inches) on all sides

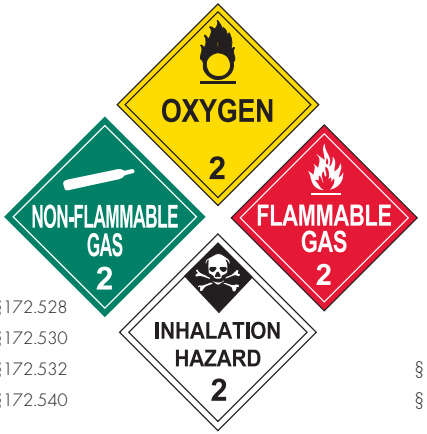
CLASS 1 Explosives



§172.522
§172.523
§172.524
§172.525

* For Divisions 1.1, 1.2, or 1.3, enter division number and compatibility group letter, when required; placard any quantity. For Divisions 1.4, 1.5, and 1.6, enter compatibility group letter, when required; placard 454 kg (1,001 lbs) or more.

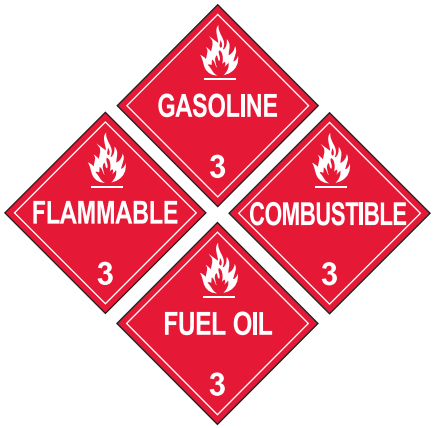
CLASS 2 Gases



§172.528
§172.530
§172.532
§172.540

For NON-FLAMMABLE GAS, OXYGEN (compressed gas or refrigerated liquid), and FLAMMABLE GAS, placard 454 kg (1,001 lbs) or more gross weight. For POISON GAS (Division 2.3), placard any quantity.

CLASS 3 Flammable Liquid and Combustible Liquid



§172.542
§172.544

For FLAMMABLE placard 454 kg (1,001 lbs) or more. GASOLINE may be used in place of FLAMMABLE placard displayed on a cargo tank or portable tank transporting gasoline by highway. Placard combustible liquid transported in bulk. See §172.504(f)(2) for use of FLAMMABLE placard in place of COMBUSTIBLE. FUEL OIL may be used in place of COMBUSTIBLE on a cargo or portable tank transporting fuel oil not classed as a flammable liquid by highway.

CLASS 4 Flammable Solid, Spontaneously Combustible, and Dangerous When Wet



§172.546, §172.547, §172.548

For FLAMMABLE SOLID and SPONTANEOUSLY COMBUSTIBLE, placard 454 kg (1,001 lbs) or more. For DANGEROUSE WHEN WET (Division 4.3), placard any quantity.

CLASS 5 Oxidizer & Organic Peroxide



§172.550, §172.552

For OXIDIZER and ORGANIC PEROXIDE (other than TYPE B, temperature controlled), placard 454 kg (1,001 lbs) or more. For ORGANIC PEROXIDE (Division 5.2), Type B, temperature controlled, placard any quantity.

CLASS 6 Poison (Toxic) and Poison Inhalation Hazard



§172.504(f)(10), §172.554, §172.555

For POISON (PGI or PGII, other than inhalation hazard) and POISON (PGIII), placard 454 kg (1,001 lbs) or more. For POISON-INHALATION HAZARD (Division 6.1), inhalation hazard only, placard any quantity.

CLASS 7 Radioactive



§172.556

Placard any quantity - packages bearing RADIOACTIVE YELLOW-III labels only. Certain low specific activity radioactive materials in "exclusive use" will not bear the label, but the radioactive placard is required for exclusive use shipments of low specific activity material and surface contaminated objects transported in accordance with §172.504(e) Table 1 and §173.427(a)(6).

CLASS 8 Corrosive



§172.558

For CORROSIVE, placard 454 kg (1,001 lbs) or more.

CLASS 9 Miscellaneous



§172.560

Not required for domestic transportation. A bulk packaging containing a Class 9 material must be marked with the appropriate ID number displayed on a Class 9 placard, an orange panel, or a white square-on-point display.

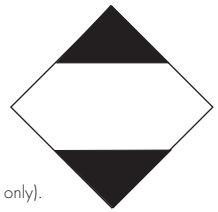
Dangerous



§172.521

A freight container, unit load device, transport vehicle, or rail car which contains non-bulk packages with two or more categories of hazardous materials that require different placards specified in Table 2 §172.504(e) may be placarded with DANGEROUS placards instead of the specific placards required for each of the materials in Table 2. However, when 1,000 kg (2,205 lbs) or more of one category of material is loaded at one loading facility, the placard specified in Table 2 must be applied.

Limited Quantity Marking



§172.315(a)(2)
(Vessel transport only).

General Guidelines on Use of Warning Labels and Placards

LABELS

See 49 CFR, Part 172, Subpart E, for complete labeling regulations.

- The Hazardous Materials Table [§172.101, Col. 6] identifies the proper label(s) for the hazardous material listed.
- Any person who offers a hazardous material for transportation MUST label the package, if required [§172.400(a)].
- Labels may be affixed to packages when not required by regulations, provided each label represents a hazard of the material contained in the package [§172.401].
- For labeling mixed or consolidated packages, see §172.404.
- The appropriate hazard class or division number must be displayed in the lower corner of a primary and subsidiary hazard label [§172.402(b)].
- For classes 1,2,3,4,5,6, and 8, text indicating a hazard (e.g., "CORROSIVE") is NOT required on a primary or subsidiary label. The label must otherwise conform to Subpart E of Part 172 [§172.405].
- Labels must be printed on or affixed to the surface of the package near the proper shipping name marking [§172.406(a)].
- When primary and subsidiary labels are required, they must be displayed next to each other [§172.406(c)].
- For a package containing a Division 6.1, PG III material, the POISON label specified in §172.430 may be modified to display the text PG III instead of POISON or TOXIC. Also see §172.405(c).
- The ORGANIC PEROXIDE label [§172.427] indicates that organic peroxides are highly flammable. Use of the ORGANIC PEROXIDE label eliminates the need for a flammable liquid subsidiary label. The color of the border must be black and the color of the flame may be black or white.

PLACARDS

See 49 CFR, Part 172, Subpart F, for complete placarding regulations.

- Each person who offers for transportation or transports any hazardous material subject to the Hazardous Materials Regulations must comply with all applicable requirements of Subpart F [§172.500].
- Placards may be displayed for a hazardous material, even when not required, if the placarding otherwise conforms to the requirements of Subpart F of Part 172 [§172.502(c)].
- For other than Class 7 or the DANGEROUS placard, text indicating a hazard (e.g., "FLAMMABLE") is not required. Text may be omitted from the OXYGEN placard only if the specific ID number is displayed on the placard [§172.519(b)(3)].
- For a placard corresponding to the primary or subsidiary hazard class of a material, the hazard class or division number must be displayed in the lower corner of the placard [§172.519(b)(4)].
- Except as otherwise provided, any bulk packaging, freight container, unit load device, transport vehicle or rail car containing any quantity of material listed in Table 1 must be placarded [§172.504].
- When the aggregate gross weight of all hazardous materials in non-bulk packages covered in Table 2 is less than 454 kg (1,001 lbs.), no placard is required on a transport vehicle or freight container when transported by highway or rail [§172.504(c)].
- Notes: See §172.504(f)(10) for placarding Division 6.1, PG III materials.
- Placarded loads require registration with USDOT. See §107.601 for registration regulations.

Placarding Tables

[§172.504(e)]

TABLE 1

Category of material (Hazard Class or division number and additional description, as appropriate)	Placard name
1.1	EXPLOSIVES 1.1
1.2	EXPLOSIVES 1.2
1.3	EXPLOSIVES 1.3
2.3	POISON GAS
4.3	DANGEROUS WHEN WET
5.2 (Organic peroxide, Type B, liquid or solid, temperature controlled)	ORGANIC PEROXIDE
6.1 (Materials poisonous by inhalation (see §171.8))	POISON INHALATION HAZARD
7 (Radioactive Yellow III label only)	RADIOACTIVE[1]

[1]RADIOACTIVE placard also required for exclusive-use shipments of low-specific-activity material and surface-contaminated objects transported in accordance with §173.427(b)(4) and (5) or (c) of the subchapter.

TABLE 2

Category of material (Hazard Class or division number and additional description, as appropriate)	Placard name
1.4	EXPLOSIVES 1.4
1.5	EXPLOSIVES 1.5
1.6	EXPLOSIVES 1.6
2.1	FLAMMABLE GAS
2.2	NON-FLAMMABLE GAS
3	FLAMMABLE
Combustible Liquid	COMBUSTIBLE
4.1	FLAMMABLE SOLID
4.2	SPONTANEOUSLY COMBUSTIBLE
5.1	OXIDIZER
5.2 (Other than organic peroxide, Type B, liquid or solid, temperature controlled)	ORGANIC PEROXIDE
6.1 (Other than materials poisonous by inhalation)	POISON
6.2	(None)
8	CORROSIVE
9	Class 9 (See §172.504(f)(9))

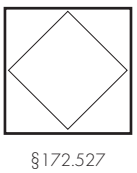
Identification Number Displays



§172.332

*Appropriate placard must be used with orange panel or white square-on-point configuration.

IDENTIFICATION NUMBER MARKINGS ON ORANGE PANELS, WHITE SQUARES-ON-POINT, OR APPROPRIATE PLACARDS MUST BE DISPLAYED ON: (1) Tank Cars, Cargo Tanks, Portable Tanks, and other Bulk Packagings; (2) Transport vehicles or freight containers containing 4,000 kg (8,820 lbs.) in non-bulk packages of only a single hazardous material having the same proper shipping name and identification number loaded at one facility and transport vehicle contains no other material, hazardous or otherwise; and (3) transport vehicles or freight containers containing 1,000 kg (2,205 lbs.) of non-bulk packages of materials poisonous by inhalation in Hazard Zone A or B. See §§172.301(a)(3), 172.313(c), 172.326, 172.328,172.330, and 172.331.



Square white background required for placard for highway- route-controlled quantity radioactive material and for rail shipment of certain explosives and poisons, and for flammable gas in a DOT 113 tank car (§172.507 and §172.510).

§172.527

For additional information contact:
The Hazardous Materials
Info Center
1-800-HMR-4922
(1-800-467-4922)
E-mail: infocntr@dot.gov
http://hazmat.dot.gov

This Chart is available on line at the following link: http://phmsa.dot.gov/hazmat

U.S. Department of Transportation
Pipeline and Hazardous Materials Safety Administration



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